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Digital transformation leadership: A public value-centered measurement scale

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ABSTRACT

We introduce a psychometrically validated scale to measure Digital Transformation Leadership in public organizations. As governments adopt new digital technologies to improve processes and services, understanding how managers effectively lead these transformations is essential. Although research on digital transformation and public sector leadership is growing, no measurement instrument grounded in a public value perspective currently exists. Following a preregistered analysis plan, we developed the scale through a rigorous multi-stage process of item development, scale construction, and evaluation. Our findings show that Digital Transformation Leadership comprises two dimensions. First, a Strategic dimension: how managers develop a vision, foster collaboration, and consider public preferences. Second, an Operational dimension: how managers guide employees, internal processes, and outputs. Additional analyses support the scale's validity, showing positive correlations with perceived leadership effectiveness, public leadership, work engagement, and job satisfaction. We further discuss how items may be used in shorter subscales or as templates. The scale offers a foundation for future research on the role of Digital Transformation Leadership in public organizations.

1. Introduction

Governments worldwide are integrating digital technologies across various policy domains and administrative functions (Criado & Gil-Garcia, 2019). These technologies can increase governments' capacity to create public value (Bannister & Connolly, 2014; Criado & Gil-Garcia, 2019). However, the increasing complexity of digital innovations also demands new approaches to governance (Andrews, 2019; Pittaway & Montazemi, 2020). Earlier waves of digitalization primarily involved transitioning from analogue to digital services to enhance efficiency and effectiveness. In contrast, the 'digital transformation' entails a fundamental redesign and reengineering of government services to meet evolving internal and external user needs (Mergel, Kattel, Lember, & McBride, 2018). For public organizations, this transformation is not just about adopting new technologies but also about fostering an organizational environment that enables change (Curtis, 2019; Gong, Yang, & Shi, 2020; Grimmelikhuijsen & Meijer, 2022). As such, the digital transformation is a comprehensive process of organizational change (Meijer, Ingrams, & Zouridis, 2022), with implications for leadership (Frick, Mirbabaie, Stieglitz, & Salomon, 2021; Schwarzmüller, Brosi, Duman, & Welpe, 2018).

While prior research has offered valuable insights, there are two remaining issues that this study aims to address. First, in addition to the many studies in the private sector, a growing body of research has examined leadership during the digital transformation in the public sector (e.g., Branderhorst & Ruijer, 2024; Mergel, Edelmann, & Haug, 2019; Nielsen, Elmholdt, & Noesgaard, 2024). Yet a comprehensive, psychometrically validated measurement instrument of Digital Transformation Leadership as it is practiced in the public sector is missing. Without a clear understanding of how public managers should lead the digital transformation, governments may struggle to advance these efforts. This paper introduces a clearly defined Digital Transformation Leadership scale, and offers a validated measurement instrument tailored to public organizations.

Second, digital transformation introduces leadership challenges related to the creation and safeguarding of public value in a legitimate and equitable manner (Andrews, 2019; Mittelstadt, Allo, Taddeo, Wachter, & Floridi, 2016). For governments, this is particularly pressing, as their legitimacy increasingly depends on whether citizens perceive changes as reliable, fair, and democratically anchored.

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Effective leadership in this context requires not only recognizing the potential of digital technologies to advance public goals, such as improving safety, sustainability, and health, but also proactively addressing ethical risks, including discrimination and threats to fundamental rights (Curtis, 2019; Sarker, Wu, & Hossin, 2018; Van der Wal, 2017). Leaders must therefore embed considerations of legitimacy into digital transformation processes. However, existing research on leadership in digital transformation has only partially captured how leaders balance innovation with the democratic, ethical, and societal responsibilities inherent to public value creation (Roman et al., 2019; Van Wart, Roman, Wang, & Liu, 2019). To address this shortcoming, this paper grounds Digital Transformation Leadership in a public value perspective, explicitly incorporating the legitimacy factor that other leadership scales have largely overlooked by including criteria such as adherence to legal and fair process requirements and ethical standards, and contributions to society.

Hence, the following research question guides this study: *How can digital transformation leadership in public organizations be conceptualized and measured from a public value perspective?* The aim of this study is to develop a psychometrically validated measurement scale for measuring Digital Transformation Leadership in public organizations. We operationalize Digital Transformation Leadership through a public value lens (Moore, 1995), drawing on Bolman and Deal's (2017) seminal work on frames of organizational change, leadership studies (Hart & Tummers, 2019; Van der Wal, 2017) and digital transformation research (Mergel et al., 2019; Schwarzmüller et al., 2018). Our intention is to address a novel and pressing empirical phenomenon, the digital transformation in the public sector, that existing measures do not sufficiently capture. At the same time, we are aware of the risk of conceptual proliferation where new concepts add limited incremental value (Banks, McCauley, Gardner, & Guler, 2016; Eva et al., 2025). By grounding our conceptualization and scale in the public value framework, we aim to provide conceptual precision and practical utility instead of just relabeling.

Following a preregistered analysis plan, we defined the domain, generated items, and validated those using focus groups with expert judges. Next, we assessed the scale in a sample of public servants in the Netherlands and conducted exploratory factor analysis for item reduction and factor extraction. Finally, we evaluated the scale by conducting dimensionality tests (confirmatory factor analysis), reliability tests, and convergent, criterion-related and discriminant validity tests.

The academic contributions of this paper are twofold. First, the paper provides a conceptual understanding and operationalization of Digital Transformation Leadership in public organizations. We contribute to the academic discourse on the digital transformation by clarifying the scale's dimensionality and studying the various behaviors necessary for guiding these processes (Branderhorst & Ruijer, 2024). This paper provides a measurement instrument that has been validated among a sample of public servants and that can be used to adequately measure Digital Transformation Leadership across public organizations in future empirical studies (Mergel et al., 2019; Pittaway & Montazemi, 2020). The dimensions identified in this scale can form the foundation for analyzing how public managers navigate this process to maximize public value and ensure legitimate implementation. Second, this paper grounds Digital Transformation Leadership in a public value perspective, which is essential for capturing what effective leadership entails in the context of public sector digital transformation (e.g., Andrews, 2019; Panagiotopoulos, Klievink, & Cordella, 2019). Importantly, the public value perspective is not treated as an abstract label but is concretely reflected in the scale items. These include, for example, items on incorporating citizen input, ensuring ethical and legal responsiveness, and delivering reliable public services.

In addition to the value of the scale for academic research, the scale provides value for public administration practice. A scale to measure Digital Transformation Leadership can provide guidance to managers who are responsible for leading in the digital transformations (Mergel et al., 2019). The scale offers a foundation for the training and personal development of managers in the public sector which may help managers to make their organizations ready for the digital age. The scale also provides an instrument for learning to take place at multiple levels, to facilitate benchmarking and learning between public organizations and between governments (e.g., Ammons & Roenigk, 2015).

2. Theory

2.1. Defining digital transformation and digital transformation leadership

Below, we discuss the definition of the digital transformation that we adopted, then turn to the concept of leadership, and emphasize the need to root Digital Transformation Leadership in a public value perspective. Finally, we propose our definition.

Over the past decade, governments have increasingly adopted digital technologies like algorithms, artificial intelligence, and big data to support internal processes and external interactions with stakeholders (Busuioc, 2021; Meijer et al., 2022). Yet scholars such as Gong et al. (2020) argue that the digital transformation is not just a matter of adopting digital technologies, as these technologies lead to 'significant changes' in, among else, infrastructure, strategies and relationships that transform public organizations. Going further, Mergel et al. (2019) even argue that scholars may use the term 'digital transformation' specifically to refer to the more cultural, organizational, and relational changes in the process of digitization and digitalization. They suggest 'digital transformation is a holistic effort to revise core processes and services of government beyond the traditional digitization efforts' (p.12). Similarly, Vial (2021, p.3) define digital transformation as 'a process that aims to improve an entity by triggering significant changes to its properties through combinations of information, computing, communication, and connectivity technologies'. A more concise definition that we will follow in this study is provided by Hanelt et al. (2021, p.1160), who recognize both the technical aspect as the broader implications by referring to the digital transformation as 'organizational change that is triggered and shaped by the widespread diffusion of digital technologies'. Following Branderhorst and Ruijer (2024), we recognize that organizational change refers not only to internal, organizational processes but also has consequences for external interactions with citizens and stakeholders.

This paper focuses on the role of leadership in and of the digital transformation. The concept of leadership has been widely studied across disciplines, with numerous definitions offered in the literature (Antonakis & Day, 2017; Kelloway & Dimoff, 2017). In the context of the public sector, Sancino and Hudson (2020, p. 703) define leadership as 'providing direction and meaning that produce followership (action)'. Similarly, 't Hart & Tummers (2019, p. 6) describe public leadership as 'a process of influencing people to think or act differently concerning public issues from what they would otherwise have done'. Effective leadership enables public sector managers to improve behavior in organizations, enhancing quality, performance, and service delivery (Hsieh et al., 2015). Additionally, public leadership involves actively scanning the environment and making sense of evolving challenges within the broader organizational and policy context (Noordegraaf, Geuijen, & Meijer, 2011; Van der Wal, 2017).

To assess what constitutes effective leadership in the context of digital transformation, we depart from a framework that clearly articulates the goals of public leadership. We argue that the creation and safeguarding of public value offers such a normative foundation. In particular, we draw on Moore's (1995) public value framework, which conceptualizes strategic leadership in the public sector through the 'strategic triangle'—comprising value-based strategy, legitimacy, and operational capacity (Bennington & Moore, 2011; Bryson, Crosby, & Bloomberg, 2014). While Moore did not originally address digital transformation, scholars have increasingly recognized its relevance for understanding how leaders manage complex, digitally driven change (e.g., Andrews, 2019; Panagiotopoulos et al., 2019).

The digital transformation of public services presents significant opportunities to generate public value, such as more efficient operations, better resource allocation, and improved service delivery (Cath, 2018; Kitchin, 2014; Meijer, 2018; Mergel, Rethemeyer, & Isett, 2016). For example, local governments use sensor technologies to monitor vehicle weight and speed, providing valuable data for infrastructure maintenance and road safety measures (Hancke & Hancke Jr, 2013). At the same time, digital transformation can also challenge core public values, raising concerns about privacy, fairness, non-discrimination, and transparency (e.g., Busuioc, 2021; Giest & Samuels, 2020; Mittelstadt et al., 2016; Ruijer, Porumbescu, Porter, & Piotrowski, 2023; Wieringa, 2023). These tensions between opportunity and risk underscore the need for value-conscious leadership in public organizations.

Drawing these threads together, scholars emphasize that public managers play a key role in navigating the value-laden challenges of digital transformation (Andrews, 2019; Mergel et al., 2019). Based on this dual lens of digital transformation and public sector leadership, we define Digital Transformation Leadership as *the process of social influence aimed at guiding organizational change towards public values that is triggered and shaped by the widespread diffusion of digital technologies, enacted by public managers in their interactions with employees, politicians, and external stakeholders*.

2.2. Current research on digital transformation leadership

Our focus on Digital Transformation Leadership in the public sector is a response to a call for both conceptual and empirical work on this topic. In the context of the digital transformation, Peng (2022) argues for a rethinking and reconceptualization of leadership. Mergel et al. (2019, p.2) point out that 'leadership need[s] to be fundamentally changed to accomplish digital transformation in the public sector.' Similarly, Gil-Garcia et al. (2018, p. 633) stress that digital transformation entails 'new styles of leadership, new decision-making processes, different ways of organizing and delivering services'. However, the conceptualization of Digital Transformation Leadership in the public sector remains relatively underexplored and is mostly influenced by private sector perspectives. Additionally, the terminology used across studies varies widely: terms such as e-leadership, digital leadership, and leadership in the digital transformation are often used interchangeably. In this study, we adopt the term Digital Transformation Leadership. While its relation to the broader concept of digital leadership is not always sharply defined, we build on Tigre, Henriques, and Curado (2025), who note that digital leadership is commonly understood as how leaders operate with digital technologies or as a blend of leadership capabilities and digital tools. We conceptualize Digital Transformation Leadership as a specific subset of digital leadership, distinguished by its narrower focus on guiding public value–laden organizational change triggered by digital technologies.

Some scholars established that the adoption of information technologies in organizations is affected by various existing leadership styles such as empowering (Frick et al., 2021) transformational (AlNuaimi, Singh, Ren, Budhwar, & Vorobyev, 2022; Ly, 2024; Shao, 2019), charismatic, transactional and servant leadership (Tseng, 2017). At the same time, like Ushaka Adie, Tate, and Valentine (2024), we believe that Digital Transformation Leadership should not be narrowly classified under a single, preexisting leadership theory. The uncritical application of established leadership paradigms without clear theoretical justification risks contributing to what Ushaka Adie et al. (2024) call 'zombie leadership': concepts that persist despite limited relevance or explanatory power.

A growing body of (mostly private sector) research has explored how leadership manifests in the context of digital transformation, resulting in a variety of conceptualizations and measurement approaches. Most frameworks do not necessarily introduce 'new' leadership behaviors, but contextualize existing ones within digital change. Yet, when critically examined, they tend to remain internally focused and fail to capture the legitimacy challenge central to public sector contexts: incorporating public preferences (input-), adhering to ethical and legal standards (throughput-), and contributing to society (output legitimacy).

Some frameworks emphasize role-based or behavioral distinctions. Weber, Krehl, and Büttgen (2022), for instance, propose a Digital Transformation Leadership Framework built around seven roles positioned along the continua of relation- versus task-orientation and external versus internal focus, ranging from the networker to the manager. While this framework does highlight external focus, it narrows this to networking and innovating, without addressing the input- or output-legitimacy of digital transformation. Tigre et al. (2025) similarly distinguish between strategic, delivery-related, interpersonal, and personal attribute dimensions. However, their focus on delivery is cast primarily in terms of employee performance.

Other studies concentrate on competency-based models. Ongena, Morsch, and Ravesteijn (2024) develop a set of digital leadership competencies tailored to business contexts, such as business planning, architecture design, and innovation. While governance is included, it remains framed in general terms of designing internal structures, not towards society. Gudergan et al. (2021) offer a comprehensive set of nine leadership dimensions linked to digital transformation success, including strategic alignment, people focus, and building trust, but their model is not operationalized as a scale and is explicitly business-oriented.

Other efforts at measurement validation further illustrate the diversity of approaches but show similar blind spots. Zahra and Wirawan (2024) present a unidimensional scale applicable across sectors, but its coverage is limited to digital knowledge, vision, and customer focus. Munsamy, Dhanpat, and Barkhuizen (2023) and Abbu, Mugge, Gudergan, Hoeborn, and Kwiatkowski (2022) adopt business-oriented perspectives, with little attention to input and output legitimacy. Bilginoğlu and Yozgat (2023) and Oktaysoy, Topcuoglu, and Kaygin (2023), in turn, focus narrowly on employees' personal leadership skills.

Finally, the European e-Competence Framework (e-CF, EN 16234–1 / CWA 16458–1:2018) formally defines a 'Digital Transformation Leader' role, emphasizing leadership in strategy, digital capability building, cultural change, and governance. Yet, this framework is not an academic measurement instrument and remains internally oriented towards organizational functioning rather than democratic legitimacy.

Taken together, existing scales and frameworks help understanding digital leadership, but they are anchored in private sector logic or general leadership theory. None explicitly adopt a public value perspective that captures the legitimacy struggles governments face in digital transformation: how leaders must balance innovation with safeguarding ethics and legal obligations, democratic accountability, and societal trust. This absence leaves a clear need for an instrument that conceptualizes and measures leadership in digital transformation as a public value practice.

Additionally, recent studies across various different countries provide important insights about leadership during the digital transformation in the public sector. These studies often focus on leadership attributes, competencies, and roles in a digital context (Zindi & Majam, 2024). For instance, Vial (2021) discusses how leadership roles in public organizations may help organizations develop a digital mindset. Amaliah and Sawitri (2023), who examine digital transformation leadership in Indonesia, highlighting dimensions such as charisma, inspirational motivation, intellectual stimulation, and individualized consideration. Similarly, Peng (2022) identifies four key attributes of 'digital leadership' in China's state governance: digital insight, digital decision-making, digital implementation, and digital guidance. Pittaway and Montazemi (2020) take a different approach by exploring the know-how required for public managers overseeing digital transformation. They define know-how as the tacit knowledge that enables managers to effectively implement technology-driven organizational processes.

Other strands of literature have also touched on aspects closely related to Digital Transformation Leadership. In the smart city context,

Sancino and Hudson (2020) analyze leadership across four modes: civic, democratic, business, and technical. Similarly, in the e-government literature, Roman et al. (2019) define and Van Wart et al. (2019) operationalize key competencies of e-leadership, including e-communication, e-social skills, e-team building, e-change management, and e-technological expertise. Hsieh, Chen, and Lo (2015) investigate leadership's role in the adoption of online services in Taiwan, concluding that in innovative environments such as e-government, leaders benefit from adopting a transformational leadership style. Finally, Hansen and Nørup (2017) show that variations in leadership during the ICT implementation process influence how well the system performs afterwards. Key actions that contribute to better outcomes include directive and participate leadership.

While above studies provide valuable insights into the attributes, competencies and roles of leadership in the digital transformation, they often do not explicitly connect leadership to the creation of public value or the ethical and legitimacy concerns that governments must address (Grimmelikhuijsen & Meijer, 2022; Rutgers & Overeem, 2014). We argue that anchoring Digital Transformation Leadership in the concept of public value offers a meaningful alternative, providing a normative and context-sensitive foundation for understanding what effective leadership entails in the digital age (Bozeman, 2007; Jørgensen & Bozeman, 2007; Moore, 1995).

The relationship between public value and emerging technologies has gained increasing attention from both scholars and practitioners in public administration (e.g., Bannister & Connolly, 2014; Twizeyimana & Andersson, 2019). However, these insights have not yet been fully integrated into discussions on digital transformation leadership. In line with Andrews (2019), we employ a public value lens to analyze Digital Transformation Leadership but, while Andrews (2019) focuses on algorithms as wicked problems, our approach considers digital transformation as a broader organizational and governance challenge.

2.3. Developing a public value perspective on digital transformation leadership

Our goal is to refine the theoretical concept of Digital Transformation Leadership by identifying key leadership dimensions that enable a successful digital transformation in the public sector. To guide this effort, we draw on Moore's (1995) public value framework. A conceptual clarification is in order here: while Moore discusses public value (singular) as a strategic management framework, he does not specify which public values (plural) should be prioritized. The term public values refers to the normative principles such as equity, transparency, accountability, or privacy that inform what governments should strive for (Bozeman, 2007; Jørgensen & Bozeman, 2007; Rose, Persson, Heeager, & Irani, 2015). Although these concepts stem from different traditions, scholars have argued they are compatible and complementary (Rutgers & Overeem, 2014). We follow this interpretation by using Moore's framework as a structure for public value creation while explicitly considering the specific public values that Digital Transformation Leadership must balance and uphold.

Following Branderhorst and Ruijer (2024), we conceptualize Digital Transformation Leadership as a higher-order construct consisting out of Moore's (1995) three dimensions strategy, legitimacy, and operational capacity. It is a reflective construct (i.e., a latent construct that is measured through multiple dimensions and subsequent indicators which act as reflection of that construct) as opposed to a formative one (which refers to a construct formed by multiple indicators) (MacKenzie, Podsakoff, & Podsakoff, 2011). Below, we further conceptualize the dimensions on the basis of key publications that connect these dimensions to the digital transformation (see Table 1 for an overview).

2.3.1. Strategy

The first dimension of Digital Transformation Leadership consists out of a value-based strategy. A value-based strategy, as defined by Moore (1995), requires public leaders to clarify strategic goals and define public value outcomes for each situation. In the context of digital transformation, this means that leaders must not only set a clear vision that establishes norms and values that support the digital transition (e.g., Andrews, 2019; Bannister & Connolly, 2014; Castelnovo, 2013; Cordella & Bonina, 2012; Twizeyimana & Andersson, 2019). This vision must be reinforced with power and status to drive meaningful change. Moreover, a value-based approach to digital transformation enables dialogue with a diverse and evolving network of stakeholders, helping leaders navigate an often ambiguous and dynamic landscape (Van der Wal, 2017).

Table 1
Dimensions and subdimensions of Digital Transformation Leadership.

(Sub)dimension	Description
Strategy	*Developing a value-based digital strategy*
Strategy: vision (SV)	Developing a shared vision on the digital transformation
Strategy: collaboration (SC)	Ensuring collaboration with stakeholders in the digital transformation
Legitimacy	*Creating a supportive environment of stakeholders*
Legitimacy: input (LI)	Ensuring public preferences (from politicians and citizens) are incorporated
Legitimacy: throughput (LT)	Safeguarding adherence to legal and fair process requirements and ethical standards
Legitimacy: output (LO)	Ensuring contribution to values important to society (public value)
Capacity	*Building an organization that can realize the strategy*
Capacity: structure (CS)	Providing the organizational structure required for realizing the digital transformation
Capacity: human resources (CH)	Managing human resources in the digital transformation
Capacity: politics (CP)	Dealing with conflicting interests in the digital transformation
Capacity: symbols (CY)	Guaranteeing the meaningfulness of work in the digital transformation

2.3.2. Legitimacy

The second dimension of Moore's (1995) strategic triangle and of our scale is legitimacy for the value-based strategy. Building upon Scharpf (1999) and Schmidt (2013), Grimmelikhuijsen and Meijer (2022) apply the concepts of input, throughput, and output legitimacy to assess the challenges of digital technologies in governance (for similar approaches, see Starke & Lünich, 2020; König & Wenzelburger, 2021; Waldman & Martin, 2022). Algorithmic decision-making, for instance, can lack democratic oversight (threatening input legitimacy), fail to meet procedural and legal standards (compromising throughput legitimacy), or produce outcomes misaligned with public values (undermining output legitimacy). Strengthening legitimacy requires integrating stakeholder preferences through democratic processes (input legitimacy), ensuring adherence to fair and legal procedures (throughput legitimacy), and aligning digital transformation outcomes with public values (output legitimacy) (Grimmelikhuijsen & Meijer, 2022).

2.3.3. Capacity

The third and final dimension in Moore's (1995) framework that is part of our scale refers to the building of operational capacity. Public leaders must develop the necessary financial, human, and technological resources (both within and beyond their organizations) to achieve desired public value outcomes (Bennington & Moore, 2011). In their comprehensive overview of perspectives on organizational science, Bolman and Deal (2017) outline four frames of organizational change. These frames form the basis for our conceptualization of capacity-building in digital transformation (see also: Meijer et al., 2022). The structural frame focuses on designing and implementing technological systems. Second, the human resources frame emphasizes understanding employees' strengths, skills, emotions, and concerns. These are critical elements in the required change of mindset and competencies among public servants (Bolman & Deal, 2017; Mergel et al., 2019). Public

managers not only need to encourage digital skills development among employees but must also cultivate their own digital expertise (Dingelstad, Borst, & Meijer, 2022; Sancino & Hudson, 2020; Streib & Navarro, 2008). Third, the political frame highlights the competitive nature of organizations, where resources are limited and interests often conflict. Digital Transformation Leadership requires an awareness of these competing interests and the ability to navigate them to achieve public value (Bolman & Deal, 2017). Finally, the symbolic frame addresses issues of meaning and organizational identity, emphasizing the importance of inspiring and motivating employees by giving purpose to the digital transformation efforts (Bolman & Deal, 2017; Meijer et al., 2022). Related to the symbolic frame, Nielsen et al. (2024) have shown how managers play a vital role in producing the narratives about the digital transformation to showcase its meaning to their organization.

Building on Moore's work, we developed a nuanced understanding of Digital Transformation Leadership in the public sector, aligning strategic vision, legitimacy, and capacity with the evolving demands of digital governance. We visualize our conceptual model in Fig. 1 and summarize our approach in Table 1. This conceptual understanding of Digital Transformation Leadership forms the basis for the development of a measurement scale.

3. Methods

Our research process, summarized in Table 2, consisted of three empirical phases with multiple sub phases (Boateng et al., 2018): item development, scale development, and scale evaluation. We preregistered our analyses on the Open Science Framework (Van Roekel, van Dam, Branderhorst, Tummers, & Meijer, 2024).

3.1. Item development

To operationalize the latent concept of Digital Transformation Leadership, we generated items based on combined deductive and inductive methods (Boateng et al., 2018). First, we based our theoretical (sub)dimensions on a qualitative study by Branderhorst and Ruijer (2024). They conducted 28 exploratory interpretative interviews with public servants, mayors / aldermen and experts to develop the concept and (sub)dimensions of Digital Transformation Leadership. Second, we conducted a narrative literature. The review was qualitative in nature

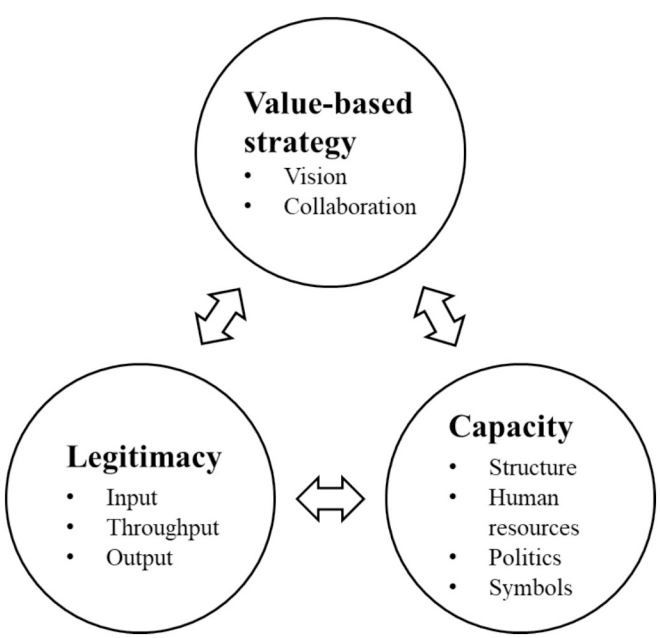

Fig. 1. Conceptual model (Branderhorst & Ruijer, 2024; Moore, 1995).

Table 2
Research process (cf. Boateng, Neilands, Frongillo, Melgar-Quiñonez, & Young, 2018).

Phase	Step	Activity	Description
1		***Item development***	
1	1	Domain identification & item generation	We defined the purpose (*develop a scale for Digital Transformation Leadership from a public value perspective*), verified there were no existing instruments, introduced a conceptual definition and dimensions. We then combined a deductive literature review with an existing inductive qualitative study (based on 28 interviews; Branderhorst & Ruijer, 2024) to generate items.
1	2	Content validity	We organized focus groups with 7 expert judges (with preceding written feedback) to validate the items.
2		***Scale development***	
2	3	Pre-test	After validation, we organized focus groups with the expert judges to assess whether the items reflected the domain of interest and produced valid measurements.
2	4	Survey administration and sample size	We administered a survey among a Flitspanel sample of public servants in the Netherlands ($N = 663$), following recommended sample sizes for both exploratory ($N = 329$) and confirmatory factor analysis ($N = 334$).
2	5	Item reduction	We conducted exploratory factor analysis with principal component factoring and oblimin rotation to reduce items based on factor loadings, inter-item and inter-total correlations.
2	6	Factor extraction	We used exploratory factor analysis to determine the optimal number of factors based on eigenvalues.
3		***Scale evaluation***	
3	7	Dimensionality tests	We conducted confirmatory factor analysis using the standard maximum likelihood estimator, allowed for some correlated error terms, and compared to alternative models, evaluating various model fit indices (i.e., CFI, TLI, RMSEA, AVE, CR). We then created scale scores for further analysis.
3	8	Reliability tests	We calculated Cronbach alpha's and hierarchical structure and inter-dimension correlations to assess scale reliability.
3	9	Validity tests	We assessed the nomological network of the scale by testing convergent validity (estimating relationships with similar scales), criterion-related (concurrent) validity (estimating relationships with concurrent outcome variables), and discriminant validity (estimating relationships with distinct scales).

and primarily based on the research team's expertise and familiarity with key theoretical frameworks and recent empirical studies in Digital Transformation Leadership. Rather than following a systematic review protocol, this approach allowed us to synthesize diverse perspectives and identify additional relevant concepts.

For the linguistic component in item generation, we considered DeVellis' (2003) recommendations for scale development, such as using simple words, avoiding double-barreled items, and avoiding double negatives. Importantly, all items described concrete, observable

behaviors (Fischer, Dietz, & Antonakis, 2024), but are general enough to be to be applicable across a wide range of public sector contexts and leadership roles. Some items refer more specifically to 'digital technologies' where this was deemed appropriate, but most items refer to a behavior related to the 'digital transformation' in general.

Next to structured discussions with four researchers, the initial item pool was independently and carefully reviewed and evaluated by multiple focus groups with experts with domain knowledge in Digital Transformation Leadership (Table 3 describes the experts). This qualitative assessment served two purposes in distinct meetings. First, it generated content validity by helping to refine the items for clarity, relevance, and coverage. Second, it served as a pre-test to assess whether the items reflected the domain of interest and produced valid measurements.

The outcome of this first phase was a set of 42 items to measure Digital Transformation Leadership. We structured these items using Branderhorst and Ruijer (2024), following their framework that distinguishes between dimensions and subdimensions of Digital Transformation Leadership (see Table 1). Example items include 'My manager communicates a coherent vision of what the digital transformation means for the organization' (dimension Strategy, subdimension Vision) and 'My manager pays attention to the ethical aspects of the digital transformation' (dimension Legitimacy; subdimension Throughput). Table 4 below shows how our operationalization process specifies the generic public value framework to a specific item.

For each subdimension, we made sure to at least develop four items. This aligns the recommendations by Hinkin (1998), who notes that at least four items are needed to test the homogeneity of items with each latent construct. In line with Hinkin (1998, p. 110), we used 5-point Likert scales, as he notes that 'it is suggested that the new items be scaled using 5-point Likert scales'. All items are shown in Appendix A.

3.2. Scale development: survey administration

After we developed the items, we tested the psychometric properties of the corresponding scale via a sample of Dutch public servants organized by Flitspanel in October/November 2024. The Flitspanel is an initiative of the Ministry of the Interior and Kingdom Relations. The initiative allows research into employment conditions, the labor market, and the working conditions of public servants in The Netherlands. All members of Flitspanel have provided their informed consent and Flitspanel follows GDPR guidelines. Their samples include public servants from most (semi-) public sectors in The Netherlands. Our target group comprises employees working in national government, provinces, municipalities, water boards (regional governing bodies tasked with managing water independently from provinces and municipalities), and joint public sector arrangements and cooperative bodies (this refers to cooperations between several provinces, municipalities or water boards). We introduced one additional criterion relevant to our study: public servants should have one or more manager(s) to be able to evaluate the items on Digital Transformation Leadership.

Around 3400 Flitspanel members belong to our target group and were invited. Of those members, 663 respondents finalized our survey, a response rate of around 19.5 %. This is sufficient for the analyses, as the sample should include at least 200–300 respondents per analysis (for exploratory and confirmatory analysis, separately) (Weston & Gore, 2006). Table 5 provides the descriptive statistics of the respondents.

Regarding sample representativeness, we note that our sample includes respondents from all relevant governmental sectors, but has an underrepresentation of municipal employees. In 2023, the distribution of all employees at the Dutch governmental levels that we include was as follows: national government: 38.69 %; provinces: 3.12 %; joint arrangements: 11.86 %; municipalities: 43.35 %; water boards: 2.98 %. Moreover, in 2023, the average age of employees across governmental sectors ranged from 45.3 (national government) to 47.8 (provinces),

Table 3
Description of independent experts.

Nr.	Job title	Organization
1	Head of Information Provision and Communication Policy	Ministry of the Interior and Kingdom Relations
2	Manager National Academy for Government Digitalization and Information	Ministry of the Interior and Kingdom Relations
3	Program Manager Digitalization	Association of Provinces of the Netherlands
4	Chief Information Officer	Province of Flevoland
5	Manager Digitalization	Association of Netherlands Municipalities
6	Manager Digital Government & Services	Association of Netherlands Municipalities
7	Strategic Manager (incl. Innovation and Digital Transformation)	The Water Authorities House

Criteria for being included as an expert were: working in one of the Dutch governmental levels (we included at least one expert from every level); and performing job tasks directly related to the digital transformation.

Table 4
Operationalization of public value: an example.

Scale →	Dimension →	Sub dimension →	Item
Public value framework (Moore, 1995)	Legitimacy (Creating a supportive environment of stakeholders; Moore, 1995)	Input legitimacy (Ensuring public preferences (from politicians and citizens) are incorporated; Grimmelikhuijsen & Meijer, 2022)	Citizen input (LI4: …ensures that citizens or other users of our services can give their opinion on the use of new digital technologies.)

Table 5
Sample characteristics ($n = 663$).

Characteristic	N	%
Gender		
Male	452	68.2
Female	207	31.2
Other	4	0.6
Sector		
National government	337	50.8
Provinces	46	6.9
Joint arrangements or cooperative bodies	40	6.0
Municipalities	190	28.7
Water boards	50	7.5
Management position		
Yes	87	13.1
No	574	86.6
Unknown	2	0.3
Amount of employees under respondents' manager		
1–5	20	3
6–10	29	4.4
11–15	61	9.2
16–20	125	18.9
21–50	300	45.2
51–100	64	9.7
>100	64	9.7
Educational level		
Primary education	11	1.7
Secondary or vocational education	128	19.3
(Applied) university or doctoral degree	522	78.7
Other	2	0.3
Age		
M (SD)	56.76 (8.04)	

indicating that our sample is somewhat older. Lastly, the proportion of women in these sectors varied from 30.7 % (water boards) to 55 % (municipalities), suggesting a slight overrepresentation of male employees in our sample (retrieved from https://www.kennisvandeoverheid.nl/cijfers-overheidspersoneel/omvang-en-samenstelling-personeelsbestand).

3.3. Scale development and evaluation

As is common for reflective constructs measuring a latent concept, our preregistered analysis plan consisted out of testing the factor structure and assessing validity. We tested the factor structure in two ways. First, we conducted exploratory factor analysis via SPSS on 329 randomly selected employees. We preregistered to use principal component factoring and oblimin rotation (we refer to extracted components as 'dimensions' in this paper). We did not specify the number of factors a priori. Next, we performed confirmatory factor analysis using R on the other 334 employees. For confirmatory factor analysis, we aimed to further test models based of the scale items including its (sub) dimensions. We preregistered to use the comparative fit index (CFI), the Tucker-Lewis index (TLI), and the root mean square error of approximation (RMSEA) to assess the fit of the models. For further analysis of the dimensions, we also mentioned to assess the square root of the average variance extracted for the dimensions (AVE) and the Composite Reliability (CR).

After establishing a final scale through exploratory and confirmatory factor analysis, we created the scale scores and calculated Cronbach alpha's and hierarchical structure and inter-dimension correlations to assess scale reliability. We then tested our measurement instrument's convergent, criterion and discriminant validity. Collectively, these analyses were guided by theoretical expectations and serve as a test of the scale's nomological validity (i.e., the place of our scale in the broader nomological network). Specifically, we analyzed the relationship of Digital Transformation Leadership roles with various other variables.

To test convergent validity, we required conceptually related constructs that should behave similarly to our scale if it functions well. First, we selected perceived leadership effectiveness (Van Knippenberg & Van Knippenberg, 2005), a widely used benchmark for evaluating whether a leadership construct captures variance in what employees broadly perceive as effective leadership. We expected that managers who score higher on Digital Transformation Leadership would also be seen as more effective leaders. Second, because the public value orientation is central to our conceptualization, we included the public leadership scale (Tummers & Knies, 2016; Vogel, Reuber, & Vogel, 2020), which explicitly captures public value creation roles. We expected that managers who score higher on Digital Transformation Leadership would also score higher on public leadership. Accordingly, we preregistered Hypothesis 1 on convergent validity, predicting that *Digital Transformation Leadership is positively related to perceived leadership effectiveness* (H1a) *and to the four public leadership roles* (H1b).

To test criterion validity, we required outcomes consistently linked to leadership and expected to be relevant in the context of the digital transformation. We selected work engagement (Schaufeli, Shimazu, Hakanen, Salanova, & De Witte, 2017) and job satisfaction (Wanous, Reichers, & Hudy, 1997) as proximal motivational and affective outcomes that are widely validated and directly linked to leadership behaviors (Skakon, Nielsen, Borg, & Guzman, 2010). They reflect employees' daily experiences of leaders' actions more closely compared to more distal or multifactorial concepts such as organizational citizenship behaviors, which are likely also shaped by broader organizational factors (Kleine, Rudolph, & Zacher, 2019). We expected that employees with managers scoring higher on Digital Transformation Leadership would also report higher levels of said outcomes. Thus, we preregistered Hypothesis 2 on criterion-related validity, stating that *Digital Transformation Leadership is positively related to work engagement* (H2a) *and job satisfaction* (H2b).

To test discriminant validity, we required constructs expected to be independent from Digital Transformation Leadership. Following earlier scale development work (Tummers & Knies, 2016), we included structural job characteristics, namely, the number of working hours per week and perceived job flexibility (1–10 scale). Our rationale was that a manager may demonstrate high or low Digital Transformation Leadership regardless of whether employees are in part-time or full-time roles or in jobs with greater or lesser flexibility. These variables were not preregistered as hypotheses but served to establish that our scale is not reducible to broader job design features.

3.3.1. Additional measures

We measured perceived leadership effectiveness via the four-item scale developed by Van Knippenberg and Van Knippenberg (2005). A sample item is 'My supervisor is very effective'. The reliability was 0.943.

Public leadership was measured using the 11-item scale developed by Vogel et al. (2020), based on the 21-item scale of Tummers and Knies (2016). A sample item was 'My supervisor encourages me and my colleagues to explain our actions to various stakeholders'. Cronbach's alpha was 0.924. The subscales also had acceptable reliabilities: 0.924 for accountability leadership, 0.884 for rule following leadership, 0.818 for political loyalty leadership, and 0.916 for network governance leadership.

Work engagement was measured using the 3-item scale Schaufeli et al. (2017) developed. A sample item is 'I am immersed in my work'. The scale's reliability was 0.807.

Job satisfaction was measured using one item; 'In general I am satisfied with my job'. We measured job satisfaction with a single-item measure as Nagy (2002, p.85) notes one that such a one-item measure 'is more efficient, is more cost-effective, contains more face validity, and is better able to measure changes in job satisfaction' (see also Wanous et al., 1997).

4. Results

4.1. Item reduction and factor extraction

We firstly executed exploratory factor analyses on 329 respondents. In four iterations, we reduced the item list from 42 to 20 items, mainly because items were too closely correlated (measuring approximately the same aspect) or loaded on multiple dimensions (measuring multiple aspects at the same time). The selection of 20 items clarifies what, according to our respondents, are the best fitting items for Digital Transformation Leadership. Appendix B contains an overview of reasons for removal for all individual items. Below, we report the statistics of the four iterations of exploratory factor analysis.

4.1.1. Iteration 1

In iteration 1, we included all 42 items. We find that some variables correlate highly ($r > 0.8$) with each other but there are no variables with very low correlations ($r < 0.3$). The correlation determinant is lower than 0.00001 (which suggests multicollinearity, although this is not a significant issue in principal component analysis so we will ignore this in following iterations), the diagonal anti-image correlations are all well above the minimum of 0.5, the KMO value is 0.97 (above the minimum of 0.5) and the Bartlett's Test is significant ($p < .001$) (this indicates that we collected enough data). In this iteration, three dimensions with eigenvalues greater than 1 were extracted, which explained 69 % of the total variance, exceeding the minimum of 60 % for scale development (Hinkin, 1998). In iteration 2, we delete items that have high correlations with each other and/or load highly on 2 or more dimensions.

4.1.2. Iteration 2

In iteration 2, we included 25 items. There are no variables with very low or high correlations ($r < 0.3$ or $r > 0.8$ (although some are close to

0.8)), the diagonal anti-image correlations are all well above the minimum of 0.5, the KMO value is 0.97 and the Bartlett's Test is significant ($p < .001$). In this iteration, two dimensions with eigenvalues greater than 1 were extracted, which explained 67 % of the total variance. However, there are still three items that load on both dimensions. In iteration 3, we remove those items and are add back two items from iteration 1 that we believe are essential and want to test again in a smaller composition of items (LT1 and LT2).

4.1.3. Iteration 3

In iteration 3, we included 24 items. There are no variables with very low or high correlations ($r < 0.3$ or $r > 0.8$ (although some are close to 0.8)), the diagonal anti-image correlations are all well above the minimum of 0.5, the KMO value is 0.97 and the Bartlett's Test is significant ($p < .001$). In this iteration, two dimensions with eigenvalues greater than 1 were extracted, which explained 68 % of the total variance. At this point, we added one more iteration in which we critically reassessed the theoretical content of the items and removed any items that have relatively high correlations (e.g., just under 0.8) and have closely related content.

4.1.4. Iteration 4

In iteration 4, we included 20 items. There are no variables with very low or high correlations ($r < 0.3$ or $r > 0.8$), the diagonal anti-image correlations are all well above the minimum of 0.5, the KMO value is 0.96 and the Bartlett's Test is significant ($p < .001$). In this iteration, two dimensions with eigenvalues greater than 1 were extracted, which explained 68 % of the total variance.

4.1.5. Final scale items

Table 6 shows the final items selected in exploratory factor analysis. We observe a two-dimensional structure that, while not identical to Moore's (1995) original triad of strategy, legitimacy, and capacity, aligns closely with its underlying logic. Specifically, dimension 1 combines items from the strategy dimension with input legitimacy, indicating that these behaviors all refer to outward-facing and future-oriented strategic behaviors. Dimension 2 clusters items related to organizational capacity alongside elements of throughput and output legitimacy, suggesting that respondents perceive managing capacity, throughput and output as coherent, everyday leadership behaviors that take place largely within the organization. This structure reveals that respondents conceptually link Moore's dimensions in ways that reflect how they experience leadership: Strategic Digital Transformation Leadership addresses external engagement and long-term vision and Operational Digital Transformation Leadership focuses more on internal processes. In that sense, the two factors identified here capture the practical essence of Moore's model, albeit in a more integrated form. Fig. 2 visualizes our conceptual model with the two dimensions in comparison to our previous model.

Table 6
Final scale items*

Item	Item wording	Comp. 1	Comp. 2
Strategic Digital Transformation Leadership			
SV1	… communicates a coherent vision of what the digital transformation means for the organization.	**0.648**	0.193
SC1	… shows that they can find expertise outside the organization for insight into the consequences of the digital transformation.	**0.836**	0.038
SC2	… collaborates with external knowledge partners to improve the digital transformation strategy.	**0.899**	−0.017
SC3	… collaborates with other governments to align the digital transformation strategy with other public sector initiatives.	**0.929**	−0.068
SC4	… collaborates with companies to incorporate private sector innovations into the digital transformation strategy.	**0.896**	−0.056
LI3	… ensures support from politicians for the actions that we take in the digital transformation.	**0.717**	0.146
LI4	… ensures that citizens or other users of our services can give their opinion on the use of new digital technologies.	**0.553**	0.265
Operational Digital Transformation Leadership			
LT1	… emphasizes that actions of employees in the digital transformation must comply with legal requirements.	−0.023	**0.804**
LT2	… pays attention to the ethical aspects of the digital transformation.	0.071	**0.756**
LO1	… tries to increase satisfaction with the services of our organization by using new digital technologies.	0.195	**0.650**
LO2	… uses digital technologies to provide more reliable public services.	0.195	**0.655**
CS1	… ensures that employees have access to new digital technologies that are necessary to do their work.	0.056	**0.687**
CS4	… encourages employees to experiment with new digital technologies.	−0.097	**0.833**
CH1	… has a good overview of the employees whose work is affected by digital transformation.	0.066	**0.767**
CH3	… discusses with employees which competencies they need for the use of new digital technologies.	−0.040	**0.872**
CH6	… ensures that employees can develop their knowledge and skills in relation to the digital transformation through training.	−0.146	**0.944**
CP1	… considers diverse perspectives on the digital transformation in the organization.	0.122	**0.792**
CP4	… can unite employees around their course in the digital transformation.	0.168	**0.711**
CY1	… explains well why employees should delve into the digital transformation.	0.062	**0.801**
CY3	… pays attention to employees' concerns about how their jobs will change as a result of the digital transformation.	−0.036	**0.876**

* Separate lists of final scale items in Dutch and English are presented in Appendix C and D.

4.2. Dimensionality tests

In the next step, we examined the scale's validity by conducting confirmatory factor analysis (CFA) in R. We conducted this analysis on the other half of our sample ($n = 334$). We used the standard maximum likelihood estimator as our data was approximately normally distributed (skewness and kurtosis scores were acceptable for all items).

Our model includes the 20 items from the exploratory factor analysis. The model fit indices indicated the following scores: CFI: 0.90 (comparative fit index; this measures how well the model fits the data compared to a baseline model; should be close to 1 with around 0.90 being acceptable), TLI: 0.88 (Tucker-Lewis index; which adjusts the CFI for complexity; should be close to 1 with around 0.90 being sufficient), and RMSEA: 0.11 (root mean square error of approximation; this estimates how well the model fits the population; should be under 0.08 to be acceptable). This model fit is suboptimal, and modification indices (indicating how well the model fit would improve if parameters such as cross-loading or error correlations were taken along) show that some items have correlated error terms. This is explainable: due to the hierarchical structure of our scale with dimensions and subdimensions, some items are more closely related than to each other than other items in the same dimension.

To address this, we specify a second model in which we allow error terms to be correlated for the five pairs of items with the highest modification index (>30) (5 out of 190 possible item pairs or 2.63 % of items). These are LT1 & LT2 (legal and ethical requirements), LO1 & LO2 (improving public service outcomes of reliability and satisfaction), SC2 & SC3 (collaboration with knowledge partners and other governments), LI3 & LI4 (support from politicians and input from citizens) and CP1 & CP4 (considering diverse perspectives and uniting employees). Adding these error terms improves the model, so it now has good fit indices: CFI: 0.96, TLI: 0.95, and RMSEA: 0.071. Additionally, AVE

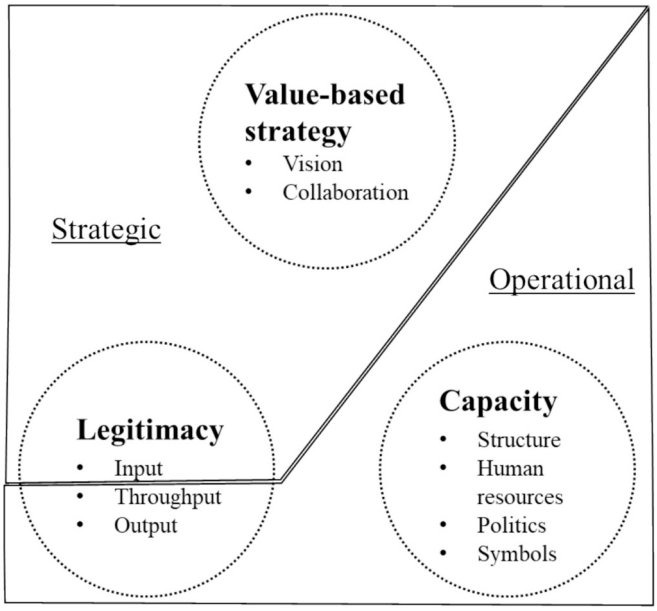

Fig. 2. New conceptual model.

(Average Variance Explained) square root values are 0.66 and 0.71 (values above 0.5 show acceptable convergent validity), and Composite Reliability (CR) values are 0.95 and 0.93 (values above 0.7 suggest good reliability).

To further test our model's fit, we compare our model to a model where all items load on one dimension. The fit of this model (CFI = 0.93; TLI: 0.91; RMSEA: 0.097) is worse than our two-factor model, which provides further evidence for its validity. A chi-square difference test was conducted to compare the two models. The two-factor model (χ^2 = 438.95, df = 164) showed a significantly better fit than the one-factor model (χ^2 = 681.42, df = 165; $\Delta\chi^2(1) = 242.47$, $p < .001$).

We also compared our two-factor model to the three-factor model based on the theoretical dimensions strategy, legitimacy and capacity. The fit of this model was slightly better than our two-factor model: CFI = 0.96; TLI: 0.96; RMSEA: 0.069. The three-factor model (χ^2 = 417.53, df = 162) showed a significantly better fit than the two-factor model (χ^2 = 438.95, df = 164; $\Delta\chi^2(1) = 21.43$, p < .001). However, the second dimension performed slightly worse on the AVE (0.75, 0.61 and 0.70) and CR (0.93, 0.86 and 0.95) compared to the scores of the two-factor model.

In this study, we chose to present the two-factor model in the main paper. Below, we report descriptives and correlations for this hierarchical structure. In addition, we present the alternative three-factor model in Appendix E for comparison. We based our decision on a combination of theoretical, empirical, and practical considerations. While our theoretical framework allows for three dimensions, the two-factor solution aligns with our EFA results, which we indicated to follow in the preregistration. Although the three-factor model shows slightly better fit indices in the CFA, the two-factor has a good fit too, and one of the three-factor model's dimensions has lower CR and AVE scores. We therefore retained the more parsimonious two-factor model in the main paper as a robust and defensible representation of the data. Nonetheless, we acknowledge that this decision is subject to interpretative discretion, which is why we offer the alternative three-factor structure in Appendix E.

Finally, we estimated a second-order factor model, yet the results showed identification issues and improper estimates (e.g., correlations above 1 between the operational dimension and the second-order factor), indicating the model is not supported by the data. The two dimensions are related but distinct.

4.3. Reliability tests

Table 7 presents the scale's descriptives and Fig. 3 presents the scale's final structure and correlations. The two dimensions are also significantly correlated ($r = 0.84$, $p < .001$). We also assessed the scale's reliability by examining Cronbach's alphas. The scale and both dimensions show sufficient reliability (>0.70). We note that the Cronbach's alpha for the full scale is very high (0.97), which may suggest there is still some overlap among items. This is partly due to the way Cronbach's alpha is calculated: it tends to increase with the number of items in a scale, as random measurement errors cancel out across items. Consequently, high alpha values are more likely in scales with many items. Furthermore, this finding may also reflect the novelty and behavioral difficulty of our scale, as respondents may not yet differentiate strongly between its various aspects. Additionally, other established leadership constructs in our study also showed high reliability. We revisit this point in the discussion.

4.4. Validity tests

Finally, we tested the nomological validity of our scale by analyzing its convergent, criterion-related, and discriminant validity (Table 8). First, there is evidence of convergent validity; our scale is positively related to perceived leadership effectiveness and public leadership. Hence, we can corroborate H1a and H1b: the relationships with similar scales are as expected.

Furthermore, on criterion-related validity, H2a and H2b are corroborated: the relationships with concurrent outcome variables are as expected. The scale is positively related to work outcomes that we hypothesized it to affect, namely work engagement and job satisfaction.

Lastly, we tested the discriminant validity of our scale with two scales that we would expect to be distinct from our scale (we chose the same scales as Tummers & Knies, 2016). For the amount of working hours, we observe that our scale is unrelated. However, for flexibility in the job, there is a positive correlation. A post hoc explanation is that Digital Transformation Leadership could be a positive factor in job flexibility, as implementing digital technologies offers employees more flexibility (e.g., by automating previously demanding tasks through AI, time is freed up for other responsibilities). Such a result indicates that flexibility was not the right scale for assessing discriminant validity.

5. Discussion and conclusions

5.1. Summary of main findings

In this study, we aimed to answer the question: *How can digital transformation leadership in public organizations be conceptualized and measured from a public value perspective?* The purpose of this study was to establish a psychometrically validated measurement scale rooted in a public value perspective to measure Digital Transformation Leadership in public organizations. To do so, we operationalized the concept of Digital Transformation Leadership and developed a measurement instrument in a rigorous and iterative process with researchers and civil servants. We tested the pool of items among a sample of Dutch civil servants and, following a preregistered analysis plan of item development, scale development and evaluation. We developed a final 20-item

Table 7
Descriptives (n = 334).

	Cronbach's alpha	M	SD
Digital Transformation Leadership	0.97	2.76	0.90
Strategic Digital Transformation Leadership	0.94	2.48	0.94
Operational Digital Transformation Leadership	0.96	2.91	0.94

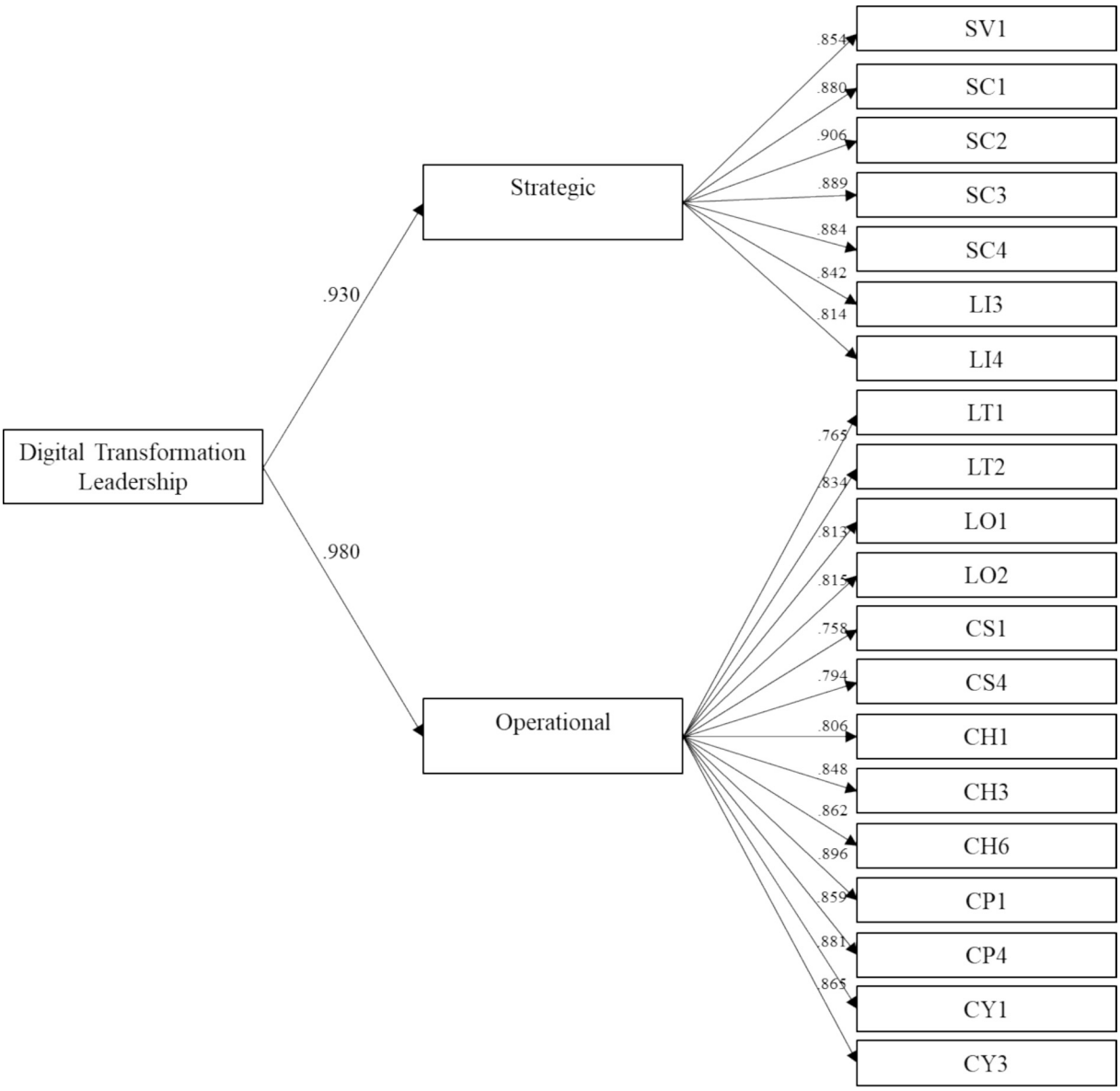

Fig. 3. Hierarchical structure scale.

Table 8
Validity tests.

Scale	Digital Transformation Leadership
Convergent validity	
Perceived leadership effectiveness	0.60***
Public leadership	0.64***
Accountability leadership	0.59***
Rule-following leadership	0.53***
Political loyalty leadership	0.31***
Network governance leadership	0.57***
Criterion-related validity	
Work engagement	0.36***
Job satisfaction	0.36***
Discriminant validity	
Amount of working hours per week	0.02
Flexibility in job	0.25***

***$p < .001$. $n = 334$.

scale that performs well, and we found that Digital Transformation Leadership encompasses two dimensions: Strategic and Operational Digital Transformation Leadership. Together these dimensions capture how public managers guide change towards public value creation in the context of the digital transformation. We further tested the scale's validity in relation to the wider nomological network and found support for our hypotheses. Using this scale, the role of Digital Transformation Leadership can be studied further. Below, we discuss implications for theory, methods, and practice, and we describe limitations of our study.

5.2. Implications for theory

The key theoretical contribution of this paper lies in the conceptualization and operationalization of Digital Transformation Leadership within public organizations. We grounded this concept in a public value perspective that captures the normative goals of public sector leadership and included key contributions such as Bolman and Deal's (2017) frames of organizational change. First, we advance conceptual clarity by bridging the literature on digital transformation in the public sector (Mergel et al., 2019) with the extensive literature on public leadership

('t Hart & Tummers, 2019). While previous studies have proposed various digital leadership competencies or dimensions—mostly rooted in business contexts or adopting a high degree of business logic (e.g., Gudergan et al., 2021; Ongena et al., 2024; Tigre et al., 2025; Weber et al., 2022; Zahra et al., 2024)—our work aims to systematically define and empirically validate the dimensions of Digital Transformation Leadership tailored to the public sector.

Second, our study adds value by grounding Digital Transformation Leadership in a public value perspective. While prior digital leadership studies in the public sector have provide many insights into the attributes and competencies of leadership within the digital transformation (e.g., Amaliah & Sawitri, 2023; Peng, 2021; Vial, 2021; Zindi & Majam, 2024) few account for the normative responsibilities that public administrations have towards citizens. We used Moore's (1995) strategic triangle as a guiding framework and enriched it using recent studies in public administration on public values and emerging technologies (e.g., Andrews, 2019; Grimmelikhuijsen & Meijer, 2022; Panagiotopoulos et al., 2019) as well as on leadership ('t Hart & Tummers, 2019; Bolman & Deal, 2017; Van der Wal, 2017) and the intersection of both themes (e.g., Frick et al., 2021; Mergel et al., 2019; Schwarzmüller et al., 2018). By anchoring our definition in Moore's (1995) public value framework, we position leadership as not only a managerial function but also a normative process aimed at realizing public value in the digital transformation. This addresses a core challenge in the public sector: determining what constitutes effective leadership amid competing values, institutional complexity, and rapid technological change. As scholars have noted (e.g., Andrews, 2019; Busuioc, 2021), the digital transformation presents both opportunities and threats to key public values such as inclusion, transparency, and fairness. Our study responds to these concerns by embedding such values into the core of our leadership framework, offering a tool for both researchers and practitioners to assess and guide leadership in a value-oriented manner.

Together, these contributions advance the scholarly debate on digital transformation and leadership by offering a theoretically grounded, psychometrically validated, and public value-aligned understanding of what effective Digital Transformation Leadership entails in public organizations.

5.3. Implications for methods

We highlight two methodological implications that we draw from this study. First, in studying the effects of leadership in the digital transformation, studies should use psychometrically sound scales instead of developing ad hoc scales. Some existing measures have limitations in terms of theoretical grounding, linguistic quality (i.e., double-barreled or ambiguous wording is used), or validity (i.e., convergent, criterion-related (concurrent), and discriminant validity are not (adequately) tested). Using such scales may lead to distorted findings and biased evidence. This paper provides an instrument that has been firmly grounded in a public value perspective, carefully formulated using best practices in scale development, and validated among a sample of public servants. Our scale can be used to adequately measure Digital Transformation Leadership in future empirical studies and increase the quality of the research conducted (DeVellis, 2003).

Second, we want to stimulate fellow scholars to use our scale creatively. For one, the strategic and operational dimensions of our scale could be used independently depending on researchers' needs. For example, scholars focusing on lower-level managers could consider only using the operational dimension. Similarly, scholars could use (sub)dimensions of items (or the alternative three-dimensional structure). For example, scholars interested in the way public managers manage human resources could use the items starting with 'CH' (Capacity: Human Resources).

Beyond this, we encourage colleagues to adapt our scale to study other transformational processes. Many of the leadership behaviors may be relevant to other societal or organizational transformations. Scholars could use our scale as a template by replacing 'digital transformation' or 'digital technologies' with another transformation or transformational component. For example, applying an item to the sustainability transformation could read: '… emphasizes that actions of employees in the sustainability transformation must comply with legal requirements'. In Appendix F we present the full template.

5.4. Implications for practice

This study has implications for public policy and management. Most obviously, public managers can use the scale in various ways. First, it can be used diagnostically to assess Digital Transformation Leadership potential within management levels and to identify specific strengths and weaknesses. For example, a manager who scores high on the Operational but low on the Strategic dimension might recognize the need to invest more in vision-building and stakeholder engagement. Second, in long-term projects, the scale can be administered periodically to track digital transformation progress over time and to examine whether interventions or reforms are accompanied by growth in leadership. Third, for training purposes, the scale can serve as a pre- and post-assessment tool to evaluate whether programs effectively enhance Strategic and/or Operational Digital Transformation Leadership, offering concrete feedback to both trainers and participants. We also suggest that the scale can be applied at the team level, where aggregated scores may help organizations explore leadership cultures within units and tailor development initiatives accordingly. Finally, across teams and organizations, the scale can facilitate benchmarking (perhaps even on an item level) and shared learning by identifying comparative strengths and weaknesses (e.g., Ammons & Roenigk, 2015).

Beyond scale application, this study also helps policymakers and managers clarify what leadership entails in the digital transformation. Despite calls for more research (e.g., Mergel et al., 2019), the notion of digital (transformation) leadership in the public sector remains under-explored and is often interpreted differently across organizations. Our study provides a clear conceptual definition and empirically validated dimensions that help specify which skills and behaviors matter for leading the digital transformation. By grounding these dimensions in public management theory, the study offers a vocabulary for diagnosing the causes of success or failure in digital transformation efforts. For example, drawing on Bolman and Deal's (2017) frames may help determine whether capacity deficits in digital transformation leadership reflect structural, human resource, political, or symbolic issues. Such analyses can also inform policy decisions about how to structure roles and responsibilities in the digital transformation, including the extent to which leadership should be (de)centralized and which responsibilities should be assigned to different organizational levels.

5.5. Limitations

As with any study, this research has limitations and should be seen as a foundational step towards developing a robust scale for measuring Digital Transformation Leadership. The validation of measurement instruments is an ongoing process (DeVellis, 2003), and future research can further assess the scale's applicability in different contexts and refine it if necessary. Similar iterative approaches have been employed in related domains, such as the development of scales for Public Service Motivation, where substantial efforts have been made to enhance measurement accuracy (beginning with Perry, 1996, followed by Vandenabeele, 2008, and most recently Kim et al., 2013), as well as in the field of public leadership, where Vogel et al. (2020) developed a short scale based on Tummers and Knies (2016).

A research design limitation is that we relied on a narrative review of the existing literature to identify relevant themes and dimensions, rather than a systematic review, which may induce some selection bias. However, to mitigate this limitation, we also grounded our item development in a published qualitative study based on 28 in-depth

interviews, ensuring that the items are not only deductive and theory-informed but also reflect the inductive perspectives of practitioners.

A few notes should be made on the scope and specificity of the items. While we drew on Moore's (1995) comprehensive public value framework to develop a broad initial pool of items, some perspectives relevant to Digital Transformation Leadership may have received less emphasis in the final scale. One such area is co-creation, which has been identified as a vital aspect in the digital transformation by recent studies (Luna, Picazo-Vela, Buyannemekh, & Luna-Reyes, 2024; Osborne, Powell, Cui, & Strokosch, 2022; Scupola & Mergel, 2022). Although our scale includes items on citizen and user input (LI4), political support (LI3), and collaborations with stakeholders (SC1-SC4), it offers a relatively limited operationalization of co-creation. Notably, our initial item pool included more targeted co-creation items, but these were not retained in the final scale. This may reflect the reality that co-creation is not yet a primary concern for many managers navigating early stages of digital transformation, an interpretation supported by the relatively low mean scores on Strategic Digital Transformation Leadership.

Besides, there is an absence of items referencing specific digital technologies or systems in the final validated scale. We interpret this not as an oversight, but as an indication that highly specific items, such as those about technologies to analyze users or the outsourcing of digital services, may be too narrow for a general leadership scale focused on broader organizational change that is only just beginning. These more specific aspects may be better suited to instruments targeting IT professionals or digital specialists, rather than public managers overseeing strategic and operational transformation.

Additionally, some of our items in our scale appear broad, yet this is a deliberate design choice that we believe reflects the reality of Digital Transformation Leadership in public organizations more. The goal of the current scale is not to capture narrow tasks, but to measure general leadership behaviors that are relevant across a wide variety of roles, organizational contexts, and stages of digital maturity. Given the diversity of public sector organizations and the evolving nature of the digital transformation, overly specific items, like those referencing particular time-bound steps in the digitalization process or specific technologies, may limit the scale's applicability and relevance. Instead, broad yet behaviorally anchored items allow for consistent measurement of the Strategic and Operational Digital Transformation Leadership dimensions.

Next, we want to discuss the broadness of, as well as potential biases in, our sample. While we recorded plenty of data from our respondents, we did not directly measure respondents' degree of involvement in digital transformation efforts. Primarily, our intention was to develop a general leadership scale applicable across public administration contexts. The digital transformation affects nearly all areas of public administration and most jobs, either directly or indirectly, which is why we aimed to capture perceptions and behaviors from a wide pool of public employees, not just those in explicitly digital transformation roles. There could be a degree of self-selection bias in the sample, where certain public employees are more likely to participate than others (e.g., those interested or with experience in the digital transformation). Future research could zoom in on specific groups by including a measure of level of involvement or interest. Besides, our data were collected in the Netherlands, with an underrepresentation of municipal employees, younger respondents, and female respondents. While we do not anticipate that this significantly affected the scale development process, replication in different contexts, such as countries and across time periods, would be valuable. Such efforts would enhance the assessment of the scale's generalizability and help identify potential boundary conditions.

Our study relies on single-source, cross-sectional survey data, which may raise concerns about common-source bias and potential inflation of correlations among variables (Favero & Bullock, 2015; Jakobsen & Jensen, 2015). However, the primary aim of this study was to develop and validate a new measurement scale, and for this purpose single-source data are both common and generally considered appropriate (e.g., Munsamy et al., 2023; Tummers & Knies, 2016; Zahra et al., 2024). Besides, in our survey design we followed procedural recommendations to counter common source bias, such as providing clear instructions, developing short and concrete items that refer to observable behaviors, and labelling all scale points (George & Pandey, 2017; Podsakoff, MacKenzie, & Podsakoff, 2012). Moreover, the observed correlation pattern provides some reassurance: if common method variance were the main driver, we would expect uniformly high correlations across all constructs. Instead, the correlations align with theoretical expectations: Digital Transformation Leadership is more strongly related to a proximal leadership benchmark (perceived leadership effectiveness, $r = 0.60$) but only moderately related to a more distal employee outcome (work engagement, $r = 0.36$). Nevertheless, all correlations used in our validity tests should be interpreted with caution and not overextended beyond their psychometric role. Future research exploring the causal or predictive power of the scale would benefit from designs that incorporate multiple time points or data sources, and more generally consider endogeneity when developing designs.

Finally, there are some limitations in our analyses. We acknowledge that designing broad items may have contributed to a high Cronbach's alpha for the scale. While such a high level of internal consistency can suggest redundancy, it may also be explained by the novelty of digital transformation leadership, where respondents may perceive its behaviors as equally challenging. Besides, the multidimensional structure that was found in both the EFA and CFA suggests that the construct is broader than the alpha suggests. Nevertheless, future studies could explore further item reduction or specification (our suggested template in Appendix F could be useful here). Likewise, while the overall model fit was acceptable (e.g., CFI = 0.96, above 0.95 is considered excellent, the RMSEA value (0.071) indicates only moderate fit (< 0.08 is considered acceptable). This suggests that although the model captures the data structure reasonably well, there is room for improvement. Future research could refine or adapt the scale items to achieve a closer fit.

Additionally, a limitation of our study lies in the choice of validation variables. While perceived leadership effectiveness, public leadership, work engagement and job satisfaction provided theoretically defensible and widely used benchmarks, alternative constructs may have offered an even closer fit to Digital Transformation Leadership. For example, entrepreneurial or innovation-oriented leadership scales might capture leaders' ability to foster change and experimentation, while outcomes such as organizational commitment or digital readiness could be used to study additional relevant consequences. Future research should extend validation efforts by including such conceptually adjacent constructs, offering a richer test of convergent and criterion-related validity. Next, choosing job flexibility to assess discriminant validity was suboptimal as it was significantly related to our scale. This is theoretically plausible as Digital Transformation Leadership may enhance flexibility through the use of digital technologies. Future research should carefully select more independent constructs (e.g., work context variables such as commute time or general psychological constructs such as life satisfaction).

6. Conclusion

The digital transformation is not just about implementing technology, it is a major challenge to the leadership capabilities of public managers. As public organizations navigate the complexities of the digital transformation, the ability of managers to guide this both strategically and operationally will be vital to their success. By introducing and validating a scale for Digital Transformation Leadership rooted in a public value perspective, this article provides researchers and practitioners with a tool to assess leadership in the digital era. Moving forward, this scale opens new avenues for research and practice, enabling public organizations to cultivate leaders who not only embrace the digital transformation but also harness it to create public value.

Data statement

The data supporting this study's findings are openly available on the Open Science Framework at https://osf.io/hrva2/?view_only=eea7a134a6b74056bd3ff56155f24fe0.

CRediT authorship contribution statement

Henrico van Roekel: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Martiene Branderhorst:** Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Conceptualization. **Lars Tummers:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Albert Meijer:** Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT-5 to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Declaration of competing interest

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Appendix A. Initial scale items*

Item	My manager…
	Strategy: Vision (SV)
SV1	**… communicates a coherent vision of what the digital transformation means for the organization.***
SV2	… presents a strategy for the steps the organization must take in relation to the digital transformation.
SV3	… links the digital transformation strategy to societal challenges.
SV4	… shows that they think carefully about the values that are important to the organization in relation to the digital transformation.
	Strategy: Collaboration (SC)
SC1	**… shows that they can find expertise outside the organization for insight into the consequences of the digital transformation.**
SC2	**… collaborates with external knowledge partners to improve the digital transformation strategy.**
SC3	**… collaborates with other governments to align the digital transformation strategy with other public sector initiatives.**
SC4	**… collaborates with companies to incorporate private sector innovations into the digital transformation strategy.**
	Legitimacy: Input (LI)
LI1	… clearly communicates the consequences of the choices that politicians have when deploying digital technologies.
LI2	… translates technical information for politicians into the consequences for citizens or other users of our services.
LI3	**… ensures support from politicians for the actions that we take in the digital transformation.**
LI4	**… ensures that citizens or other users of our services can give their opinion on the use of new digital technologies.**
	Legitimacy: Throughput (LT)
LT1	**… emphasizes that actions of employees in the digital transformation must comply with legal requirements.**
LT2	**… pays attention to the ethical aspects of the digital transformation.**
LT3	… leads the discussion on the question of whether regulation of new digital technologies in the organization is necessary.
LT4	… critically discusses the possibilities that new digital technologies offer to analyze users of our services.
LT5	… discusses whether the collected data can be used in decision-making.
	Legitimacy: Output (LO)
LO1	**… tries to increase satisfaction with the services of our organization by using new digital technologies.**
LO2	**… uses digital technologies to provide more reliable public services.**
LO3	… increases awareness of employees of the importance of the privacy of citizens or other users of our services.
LO4	… stimulates transparency towards citizens or other users of our services about what the organization does with their data.
LO5	… increases the digital resilience of citizens or other users of our services.
LO6	… provides support to citizens or other users of our services who lack access to digital resources.
	Capacity: Structure (CS)
CS1	**… ensures that employees have access to new digital technologies that are necessary to do their work.**
CS2	… ensures that the data remains the property of the organization when outsourcing digital services.
CS3	… monitors the use of new digital technologies in the organization.
CS4	**… encourages employees to experiment with new digital technologies.**
CS5	… ensures that the lessons from pilots on new digital technologies are translated to the organization.

(*continued on next page*)

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Item	My manager…
	Capacity: Human Resources (CH)
CH1	**… has a good overview of the employees whose work is affected by digital transformation.**
CH2	… ensures that there is sufficient knowledge within the organization to manage external parties that provide digital services on behalf of the organization.
CH3	**… discusses with employees which competencies they need for the use of new digital technologies.**
CH4	… motivates employees to immerse themselves into new digital technologies.
CH5	… supports employees whose jobs will change as a result of the digital transformation.
CH6	**… ensures that employees can develop their knowledge and skills in relation to the digital transformation through training.**
	Capacity: Politics (CP)
CP1	**… considers diverse perspectives on the digital transformation in the organization.**
CP2	… knows how to create support for their course in the digital transformation in the middle of conflicting interests.
CP3	… shows the ability to steer the digital transformation effectively in a complex environment of competing interests.
CP4	**… can unite employees around their course in the digital transformation.**
	Capacity: Symbols (CY)
CY1	**… explains well why employees should delve into the digital transformation.**
CY2	… inspires employees to get started with new digital technologies.
CY3	**… pays attention to employees' concerns about how their jobs will change as a result of the digital transformation.**
CY4	… pays attention to whether employees continue to experience their jobs as meaningful in the digital transformation.

* Final scale items in bold.

Appendix B. Item assessment in EFA

Item	Removed in iteration	Reason for removal
SV1	–	–
SV2	1	High correlation with SV1 (R > 0.8)
SV3	1	High correlation with SV1 (R > 0.8)
SV4	2	Positive cross-loading on multiple dimensions (> 0.3)
SC1	–	–
SC2	–	–
SC3	–	–
SC4	–	–
LI1	1	Opposite-signed cross-loading on multiple dimensions (> 0.3)
LI2	1	High correlation with LI3 (R > 0.8)
LI3	–	–
LI4	–	–
LT1*	–	–
LT2*	–	–
LT3	1	Opposite-signed cross-loading on multiple dimensions (> 0.3)
LT4	1	Opposite-signed cross-loading on multiple dimensions (> 0.3)
LT5	3	Too specific and generally fits in LT1 and LT2
LO1	–	–
LO2	–	–
LO3	1	Negative loading on one dimension (> 0.3)
LO4	1	Negative loading on one dimension (> 0.3)
LO5	1	Opposite-signed cross-loading on multiple dimensions (> 0.3)
LO6	1	Opposite-signed cross-loading on multiple dimensions (> 0.3)
CS1	–	–
CS2	2	Positive cross-loading on multiple dimensions (> 0.3)
CS3	2	Positive cross-loading on multiple dimensions (> 0.3)
CS4	–	–
CS5	1	Positive cross-loading on multiple dimensions (> 0.3)
CH1	–	–
CH2	3	Worst loading item in dimension 1 and partially cross-loading
CH3	–	–
CH4	1	High correlation with CH3 (R > 0.8)
CH5	3	Content is also reflected in CH1, CH3 and CH6
CH6	–	–
CP1	–	–
CP2	1	High correlations with CP1 and CP4 (R > 0.8)
CP3	1	High correlations with CP1 and CP4 (R > 0.8)
CP4	–	–
CY1	–	–
CY2	3	High correlation with CY1 (R close to 0.8)
CY3	–	–
CY4	1	High correlation with CY3 (R > 0.8)

* Items LT1 and LT2 were temporarily removed in iteration 1 for cross-loading but added back in iteration 2.

Appendix C. Final scale items (Dutch)

Nr	Mijn leidinggevende…
	Strategisch digitale transformatie leiderschap
1	… communiceert een samenhangende visie over wat de digitale transformatie voor de organisatie betekent.
2	… laat zien buiten de organisatie expertise te kunnen vinden voor inzicht in de gevolgen van de digitale transformatie.
3	… werkt samen met externe kennispartners om de strategie voor digitale transformatie te verbeteren.
4	… werkt samen met andere overheden om de strategie voor de digitale transformatie af te stemmen op andere initiatieven in de publieke sector.
5	… werkt samen met bedrijven om innovaties uit de private sector mee te nemen in de strategie voor digitale transformatie.
6	… zorgt voor draagvlak van politici voor de acties die we ondernemen in de digitale transformatie.
7	… zorgt ervoor dat burgers of andere gebruikers van onze diensten hun mening kunnen geven over de inzet van nieuwe digitale technologieën.
	Operationeel digitale transformatie leiderschap
8	… benadrukt dat het handelen van medewerkers in de digitale transformatie aan wettelijke eisen moet voldoen.
9	… heeft aandacht voor de ethische aspecten van de digitale transformatie.
10	… probeert de tevredenheid over de diensten van onze organisatie te vergroten door de inzet van nieuwe digitale technologieën.
11	… maakt gebruik van digitale technologieën om betrouwbaardere publieke diensten te bieden.
12	… zorgt ervoor dat medewerkers toegang hebben tot nieuwe digitale technologieën die nodig zijn om hun werk te kunnen doen.
13	… moedigt medewerkers aan om te experimenteren met nieuwe digitale technologieën.
14	… heeft een goed overzicht van de medewerkers wiens werk beïnvloed wordt door de digitale transformatie.
15	… bespreekt met medewerkers welke competenties zij nodig hebben voor de inzet van nieuwe digitale technologieën.
16	… zorgt ervoor dat medewerkers door opleidingen hun kennis en vaardigheden in relatie tot de digitale transformatie kunnen ontwikkelen.
17	… heeft aandacht voor verschillende perspectieven op de digitale transformatie in de organisatie.
18	… kan medewerkers verenigen rondom diens koers in de digitale transformatie.
19	… legt goed uit waarom medewerkers zich moeten verdiepen in de digitale transformatie.
20	… besteedt aandacht aan de zorgen van medewerkers over de manier waarop hun baan gaat veranderen als gevolg van de digitale transformatie.

Appendix D. Final scale items (English)

Nr	My manager…
	Strategic Digital Transformation Leadership
1	… communicates a coherent vision of what the digital transformation means for the organization.
2	… shows that they can find expertise outside the organization for insight into the consequences of the digital transformation.
3	… collaborates with external knowledge partners to improve the digital transformation strategy.
4	… collaborates with other governments to align the digital transformation strategy with other public sector initiatives.
5	… collaborates with companies to incorporate private sector innovations into the digital transformation strategy.
6	… ensures support from politicians for the actions that we take in the digital transformation.
7	… ensures that citizens or other users of our services can give their opinion on the use of new digital technologies.
	Operational Digital Transformation Leadership
8	… emphasizes that actions of employees in the digital transformation must comply with legal requirements.
9	… pays attention to the ethical aspects of the digital transformation.
10	… tries to increase satisfaction with the services of our organization by using new digital technologies.
11	… uses digital technologies to provide more reliable public services.
12	… ensures that employees have access to new digital technologies that are necessary to do their work.
13	… encourages employees to experiment with new digital technologies.
14	… has a good overview of the employees whose work is affected by digital transformation.
15	… discusses with employees which competencies they need for the use of new digital technologies.
16	… ensures that employees can develop their knowledge and skills in relation to the digital transformation through training.
17	… considers diverse perspectives on the digital transformation in the organization.
18	… can unite employees around their course in the digital transformation.
19	… explains well why employees should delve into the digital transformation.
20	… pays attention to employees' concerns about how their jobs will change as a result of the digital transformation.

Appendix E. Three factor structure

Table E1 compares the scale's descriptives (full scale and two-factor structure) as presented in the main paper to those of the alternative three-factor structure. Fig. E1 visualizes the three-factor structure and presents its correlations.

The three dimensions of the three-factor structure are significantly correlated. For value-based strategy and legitimacy, $r = 0.83$ ($p < .001$), for value-based strategy and capacity, $r = 0.80$ ($p < .001$), and for legitimacy and capacity, $r = 0.84$ ($p < .001$). We also assessed the scale's reliability by examining Cronbach's alphas. The dimensions show sufficient reliability (>0.70).

Table E1: Descriptives ($n = 334$).

	Cronbach's alpha	M	SD
Full scale			
Digital Transformation Leadership	0.97	2.76	0.90
Two-factor structure			
Strategic Digital Transformation Leadership	0.94	2.48	0.94

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(*continued*)

	Cronbach's alpha	M	SD
Operational Digital Transformation Leadership	0.96	2.91	0.94
Three-factor structure			
Value-based strategy dimension	0.94	2.53	0.99
Legitimacy dimension	0.92	2.74	0.94
Capacity dimension	0.95	2.91	0.95

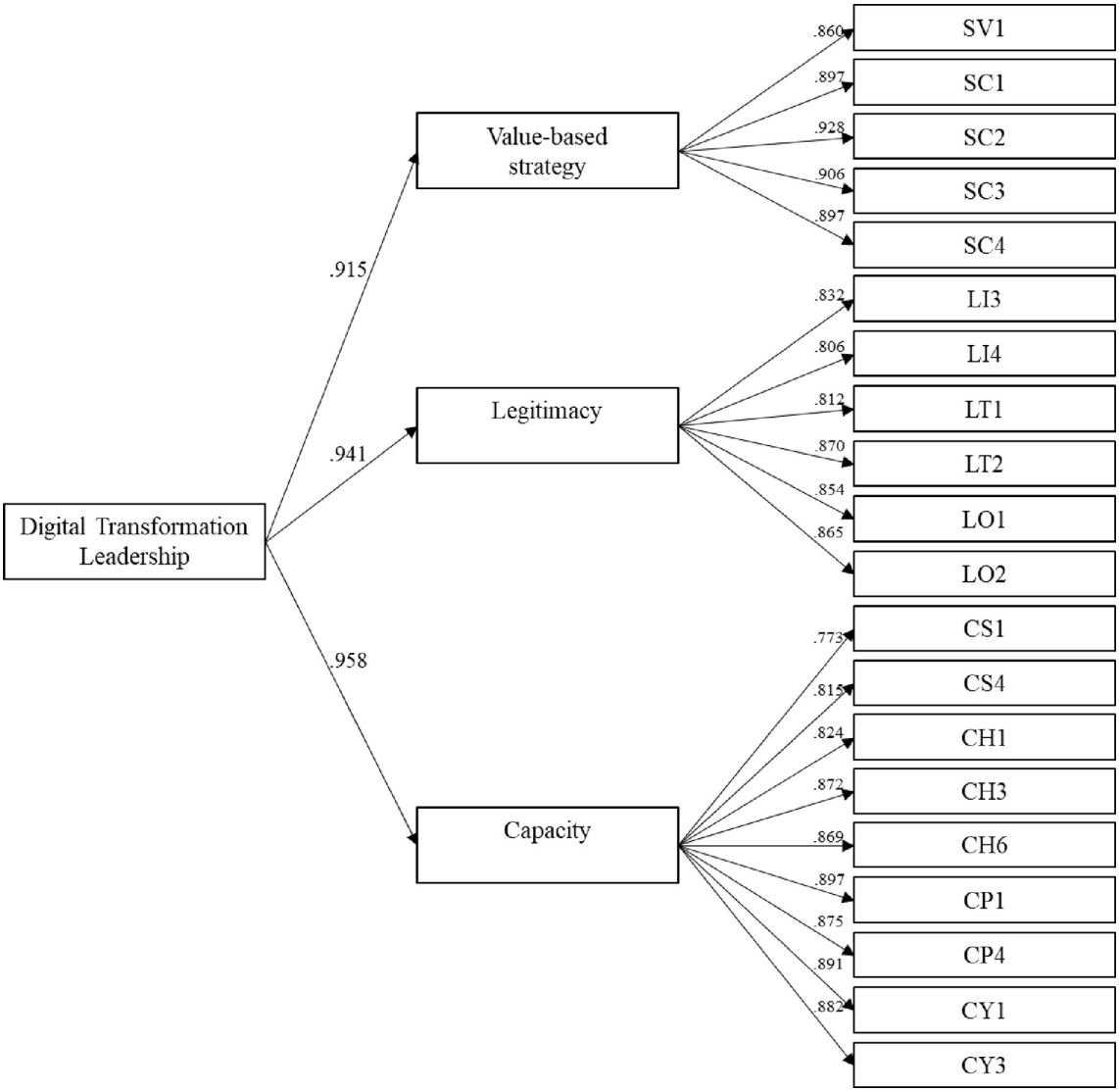

Fig. E1: Hierarchical structure three-factor scale

Appendix F. Leadership in a transformation template

Table F1: Leadership in a transformation: a template.

Strategic Digital Transformation Leadership
… communicates a coherent vision of what the *[transformation]* means for the organization.
… shows that they can find expertise outside the organization for insight into the consequences of the *[transformation]*.
… collaborates with external knowledge partners to improve the *[transformation]* strategy.
… collaborates with other governments to align the *[transformation]* strategy with other public sector initiatives.
… collaborates with companies to incorporate private sector innovations into the *[transformation]* strategy.
… ensures support from politicians for the actions that we take in the *[transformation]*.
… ensures that citizens or other users of our services can give their opinion on *[transformational component]*.

(*continued on next page*)

(*continued*)

Operational Digital Transformation Leadership
… emphasizes that actions of employees in the *[transformation]* must comply with legal requirements.
… pays attention to the ethical aspects of the *[transformation]*.
… tries to increase satisfaction with the services of our organization by using *[transformational component]*.
… uses *[transformational component]* to provide more reliable public services.
… ensures that employees have access to *[transformational component]* that are necessary to do their work.
… encourages employees to experiment with *[transformational component]*.
… has a good overview of the employees whose work is affected by *[transformation]*.
… discusses with employees which competencies they need for *[transformational component]*.
… ensures that employees can develop their knowledge and skills in relation to the *[transformation]* through training.
… considers diverse perspectives on the *[transformation]* in the organization.
… can unite employees around their course in the *[transformation]*.
… explains well why employees should delve into the *[transformation]*.
… pays attention to employees' concerns about how their jobs will change as a result of the *[transformation]*.

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